

LESSON 3-1 • SESSION 2 • PRACTICE • FORM B

Ratios and Tape Diagrams

Learning target – I can use a tape diagram to share a total according to a given ratio. · 6.RP.A.1

REMEMBER

- A **tape diagram** uses equal boxes. The ratio tells how many boxes each group gets.
- **One box = total ÷ total parts.** Multiply one box by each term.
- If you know one group, divide it by its boxes to find one box.

WORKED EXAMPLE

Ana and Ben share 28 stickers in the ratio 3 : 4. Find each share.

Step 1 Total parts: $3 + 4 = 7$.

Step 2 One box: $28 \div 7 = 4$.

Step 3 Ana $3 \times 4 = 12$; Ben $4 \times 4 = 16$. Check: $12 + 16 = 28$.

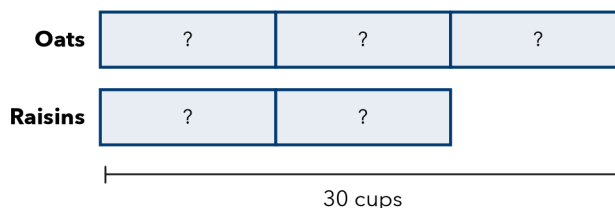
Answer Ana 12, Ben 16

WORDS tape diagram (*diagrama de cinta*) · unit (*unidad*) · total parts (*partes totales*)

PRACTICE

show your work · label units

1. **Trail Mix.** A recipe uses 3 cups of oats for every 2 cups of raisins. A chef makes a 30-cup batch.



- a. Total parts: _____
- b. Value of one box: _____ cups
- c. Oats and raisins: _____
2. **Smoothie.** A smoothie uses 4 strawberries for every 3 bananas. A pitcher uses 21 fruits. How many bananas?
3. **Mural Paint.** Art club mixes 4 cups blue with 1 cup white. They need 40 cups in all. How much white?

Hint Total parts: $4 + 3$. Divide 21 by it.

Show your work.

Answer _____ bananas

Show your work.

Answer _____ cups

4. **Fish Tank.** A tank has goldfish and guppies in the ratio 3 to 5. There are 40 fish in all. How many guppies?
5. **Camp Counselors.** A camp has 1 counselor for every 12 campers. There are 60 campers. How many counselors are needed?

Hint One box = $40 \div 8$, not $40 \div 5$.

Show your work.

Answer _____ guppies

Hint Each counselor goes with a group of 12.

Show your work.

Answer _____ counselors

6. **Splitting Money.** Two cousins share \$56 in the ratio 3 : 4.

Hint Add the terms. Divide \$56 by that total.

- a. Cousin 1: \$ ____ b. Cousin 2: \$ ____

7. **Find the Total.** A snack mix uses 3 parts raisins to 4 parts nuts. The recipe uses 20 ounces of nuts.

Hint Nuts are 4 boxes. Find one box first.

- a. How many ounces is one part? ____ oz
b. How many ounces does the recipe make in all? ____ oz

Show your work.

8. **Garden Beds.** A gardener plants carrots and peas in the ratio 7 : 5 with 96 seeds in all.

- a. Carrot seeds and pea seeds: ____
b. How many more carrot seeds than pea seeds? ____
c. Next year she wants equal amounts. What ratio? ____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Diego's Work.** A pot of chili uses 3 cups rice and 4 cups beans, 42 cups in all. Diego divided 42 by 3 and got 14 cups of rice. Explain his mistake and find the rice.

TEST PRACTICE

choose the best answer

10. A ratio of 3 : 4 shares a total of 28. One box is worth –

A 3 B 4 C 7 D 12

11. A ratio of 1 : 5 shares 30 cookies. The smaller share is –

A 5 B 6 C 25 D 29

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

16 D	9 I	35 E	5 D	15 X	720 Q	8 V	20 Z	6 D	25 I
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#1b	#2	#3	#4	#5	#7b	#8b
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____